

Designing AI to Work WITH or FOR People?

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ABSTRACT

Artificial Intelligence (AI) can refer to the machine learning algorithms and the automation applications built on top of these algorithms. Human-computer interaction (HCI) researchers have studied these AI applications and suggested various Human-Centered AI (HCAI) principles for an explainable, safe, reliable, and trustworthy interaction experience. While some designers believe that computers should be supertools and active appliances, others believe that these latest AI systems can be collaborators. With today's AI algorithm breakthroughs, in this panel we ask whether the supertool or the collaboration metaphors best support work and play? How can we design AI systems to work best with people or for people? What does it take to get there? This panel will bring together panelists with diverse backgrounds to engage the audience through the discussion of their shared or diverging visions on the future of human-AI interaction design.

CCS CONCEPTS

• **Human-centered computing** → **Empirical studies in HCI**.

KEYWORDS

human AI Collaboration; collaborative AI; AI partner; explainable AI; trusted AI; group collaboration; computer-supported cooperative work

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1 INTRODUCTION

The recent development of AI research (e.g., deep learning, neural network, and knowledge graph) [17] have influenced many

application domains to have more AI flavor. For example, AI-powered healthcare applications are extensively studied and some AI systems can perform as well as human doctors in reading medical image data [2, 10]. In addition to the healthcare domain, AI also presents enormous opportunities for the people who may interact with it [5, 28]. Drivers can dedicate the tedious parking or driving tasks to an autonomous car system [15]; Children can read storybooks together with a conversational agent in a smart speaker [29]. These are all powerful examples that illustrate the benefits that AI has brought to people in recent years.

To better support the design of user interaction with these AI applications, various researchers and design practitioners have proposed detailed design guidelines for AI (“Human-Centered AI design” or “human-in-the-loop AI design”) [1, 22]. These design guidelines often aim for higher user satisfaction, user adoption, productivity, or user engagement [21]. However, interaction is not the same as collaboration. Collaboration involves mutual goal understanding, preemptive task co-management and shared progress tracking [20]. It is one level of complexity to design for the better user experience of a conversational agent to support a user to turn on a light; it is quite different design consideration if the conversational agent is to work together with parents for early child education scenarios [29].

To fully realize the potential of AI in many scenarios, some researchers suggest designers consider the collaboration perspective in AI system design, for two reasons:

Firstly, when a computer system becomes “smart”, human users tend to mindlessly apply social rules, expectations, or even assign personalities to the system — it is known as “Computer As Social Actors” (CASA) or Anthropomorphism [19]. Thus, when humans work together with a smart AI system towards a common goal, they may treat AI as a collaborative partner instead of a tool [12]. If the AI partner does not respect interpersonal collaboration norms, it may fail (as any unsuccessful human-human collaborations) [11].

Secondly, most of human activities today are already done collaboratively [20]. Thus, to integrate AI into the already-complicated human team workflow, it is critical to translate the human collaboration norms and context into the AI system design (a “translational science” approach [3, 4], so that it should fit or complement to the established human workflow, instead of disrupt it).

However, collaboration WITH machines may be a misleading design strategy; tools that work FOR people may lead to better outcomes. Existing technologies are designed to support user control over predictable machines, which users treat as tools, tele-operated

*Dakuo Wang is the moderator. Panelists are listed in alphabetical order.

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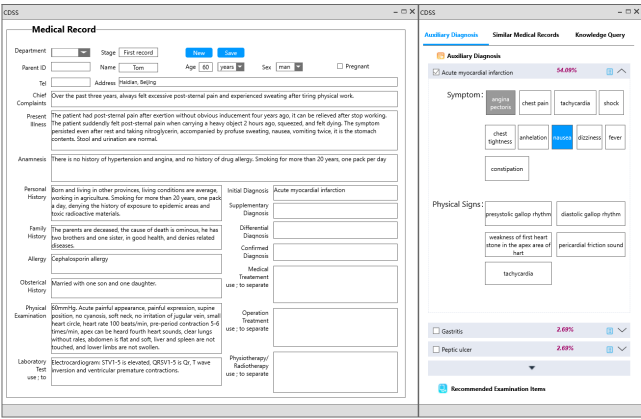


Figure 1: The screenshot of a typical AI-CDSS system user interface design: the EHR system user interface is to the left and the AI-CDSS functionality is to the right.

devices, or powerful appliances. Successful and satisfied users of digital cameras, GPS navigation systems, and most mobile device apps do not see their devices as collaborators, partners, or teammates. Future supertools and active appliances, which users control, steer, and operate, may be more effective designs than pseudo-collaborators.

The “human-AI collaboration” paradigm (or “collaborative AI design”) [9, 13, 16, 25] is not a new concept. Along with the previous AI “summers”[12], research pioneers have proposed various conceptual frameworks towards the human-AI collaboration future: Licklider originated the man-machine symbiosis computing vision [18], Nass discussed the CASA framework [19], Horvitz proposed mixed-initiative user interface design [14], and the recent human-centered AI design research [22]. One notable event was the 1997 debate between Pattie Maes and Ben Shneiderman about whether smart agency (i.e. interaction with AI) or direct manipulation would be the dominant user interface design [23]. But these design considerations were constrained by the limitations of AI algorithm in the early days.

In the past decades, the development of deep learning, faster hardware, and big data storage infrastructures could have changed the design of AI applications in various fields. Some examples are: AI-powered Clinical Decision Support systems (AI-CDSS) to work with doctors in clinics (Fig. 1) [27], conversational agent systems to play or teach with children (Fig. 2) [29], and even automated machine learning (AutoAI) system to build AI models together with data scientists (Fig. 3) [26]. But tragic failures, biased outcomes, and suboptimal performance (e.g., [24]) are also reported with some of today’s AI systems. In many cases, the AI system designers fail to translate the rich understanding of people, their complex workflow, context, and norms into the design and implementation of the system [27]. Are we already in an era of computing that an AI system serves as a partner instead of a tool [12]? What design metaphors will best serve future designers: collaboration, supertool, partnership, active appliance, teammate, direct manipulation, symbiosis, or harmony?

In this panel, researchers and practitioners with diverse backgrounds (e.g., research interest, culture, occupation, etc.) will discuss

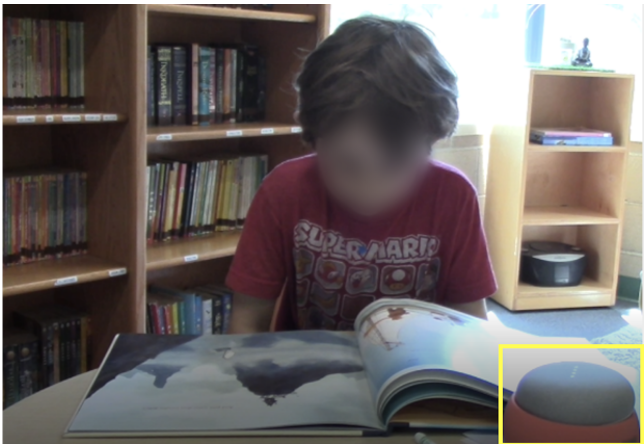


Figure 2: A smart speaker system reads a storybook to a child and interactively asks questions and receives answers during the process, as reported in [29].

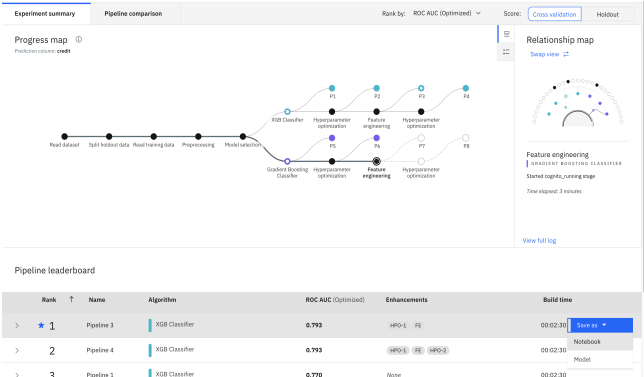


Figure 3: The screenshot of an AutoAI/AutoML system user interface design reported in [26]: the UI shows a result visualization at the top and a model leaderboard at the bottom that human data scientists and AI automation work together to train AI models.

how they think of today’s AI, in comparison to the previous waves of AI techniques. How does AI change how we work, play, and live, and what it means to collaborate with an AI or use a supertool. We hope that the panelists’ discussion and audience feedback will help bridge the gaps among different communities (e.g., HCI and AI, practitioner and researcher, Euro-western and emerging markets) to pursue this research agenda.

We will start by exploring the current challenges and opportunities for the HCI and AI research community. For example:

- What is considered to be an AI system in comparison to a non-AI computer system?
- Are we already in the computing era of human-AI collaboration?
- Should the collaboration with AI be the same or different from the collaboration with human partners?

- Does the modality make a difference (e.g., physical robot, text chatbot, spoken dialog, etc.). If so, how?
- What are appropriate roles for AI systems: leader? facilitator? participant? advisor? How do those roles affect the dynamics of human group interaction?
- Does AI need “collaboration awareness” [6] for human emotions, conventions and mental models?
- Since trust is crucial for collaboration (e.g., [7]), how will AI affect trust within a hybrid human-AI teaming?
- What ethical considerations are there for AI collaborators?
- Should AI be trained to withhold information, to be diplomatic, to be deceptive if necessary for the social good, or to be completely transparent [8]?
- How does responsibility for failures govern collaboration between humans and machines?

Then, we will ask design and research questions, such as:

- What are the good and bad examples of AI systems that work together with humans today?
- What theories in HCI and CSCW can shed light on the AI system design or research on Human-Centered AI?
- Which metaphors are appropriate for portraying AI? How should agents be depicted in terms of team roles?
- What are the current development and user testing practices for AI that are already in the real world?

2 PANEL MODERATOR

Dakuo Wang, Research Team Lead, IBM Research, USA

Dakuo’s research lies in the intersection between human-computer interaction (HCI) and artificial intelligence (AI). He is now leading a team of researchers, engineers, and designers to conduct research and design user experience for various types of AI applications, including automated machine learning [26], chatbots [29], and clinical decision support systems (CDSS) systems [27]. From building and researching these AI systems in different domains, he aims to triangulate the design paradigm for supporting “human-AI collaboration”. Dakuo now serves as ACM CSCW’22 Sponsor Co-Chair, ACM CHI’22 SIG Co-Chair, and AAAI ICWSM’21 Inclusion and Diversity Co-Chair.

3 PANELISTS INVITED

The panelists are top researchers from the CHI and AI community, experts in collaboration and/or AI. Panelists hold a variety of viewpoints about the intelligence and capabilities of AI systems. Some think AI are far from smart and never will be. Other panelists have spent many years designing interaction of AI to make it acceptable to consumers. Our panelists also come from different countries and professions. Brief biographical sketches and proposed panel statement topics for each panelist appear in alphabetical order below:

Pattie Maes, Professor, MIT, USA

Pattie’s position statement: AI systems will only succeed in living up to their potential if their design is centered around people. For almost 30 years now, I have focused on designing IA (intelligence augmentation) as opposed to AI systems. I plan to talk about the lessons learned and principles developed over those years. Pattie runs the Media Lab’s Fluid Interfaces research group, which

aims to radically reinvent the human-machine experience. Coming from a background in artificial intelligence and human-computer interaction, she is particularly interested in the topic of cognitive enhancement, or how immersive and wearable systems can actively assist people with cognitive skills and functions such as memory, attention, motivation, learning, decision making, communication, and wellbeing. In 1997, Pattie Maes and Ben Shneiderman had a famous debate about whether the smart agency (i.e., AI) or direct manipulation would dominate user interaction design [23].

Xiangshi Ren, Professor, Kochi University of Technology, Japan

Xiangshi’s position statement: The new goals of “Human-AI Collaboration” could be “Human-Engaged AI” through a Human-Engaged Computing Perspective [21].

Xiangshi REN is a lifetime tenured professor in the School of Information and founding director of the Center for Human-Engaged Computing (CHEC) at Kochi University of Technology. He was a visiting professor at the University of Toronto, visiting faculty researcher at IBM Research (Almaden), and visiting/guest/chair professor at several universities in China. He has been working on fundamental studies in the field of Human-Computer Interaction (HCI) for 30 years. In recent years, he and his colleagues have established a new conceptual framework, Human-Engaged Computing (HEC), for rethinking the relationship between humans and computers which refers to the synergism between human capacities and technological capabilities. He has received best paper awards from various conferences and journals such as ACM CHI, ISS, IEEE SmartComp, CCF, IPSJ. He is founding president and honorary lifetime president of the International Chinese Association of Computer Human Interaction (ICACHI). He is a Senior Member of the ACM, Senior Member of the IEEE, a Fellow of IPSJ, and a Member of the Engineering Academy of Japan.

Ben Shneiderman, Emeritus Distinguished University Professor, University of Maryland, USA

Ben’s position statement: The new goals of Human-Centered AI are to produce designs that are Trusted, Reliable & Safe (TRS) [22]. Achieving these goals will dramatically increase human performance, while supporting human self-efficacy, mastery, creativity, and responsibility, while also respecting human values, justice, and dignity.

BEN SHNEIDERMAN is an Emeritus Distinguished University Professor in the Department of Computer Science, Founding Director (1983-2000) of the Human-Computer Interaction Laboratory, and a Member of the UM Institute for Advanced Computer Studies (UMIACS) at the University of Maryland. He is a Fellow of the AAAS, ACM, IEEE, and NAI, and a Member of the National Academy of Engineering, in recognition of his pioneering contributions to human-computer interaction and information visualization. He has received 6 honorary doctorates for his contributions, which include the clickable highlighted web-links, high-precision touch-screen keyboards for mobile devices, and tagging for photos. Shneiderman’s information visualization innovations include dynamic query sliders for Spotfire, development of treemaps for viewing hierarchical data, novel network visualizations for NodeXL, and event sequence analysis for electronic health records.

Yuanchun Shi, Professor, Tsinghua University, China

Yuanchun's position statement: Identify users' natural input intention from the continuous and random data should combine AI and HCI technologies into an end-to-end decoding framework.

Yuanchun Shi is a Changjiang Distinguished Professor in the Department of Computer Science and Technology, Tsinghua University, China. She received her PhD, MSc and BSc from Tsinghua, and joined its faculty in 1993. Her research in human-computer interaction, ubiquitous computing, and multimedia has won many academic awards, including two State Science and Technology Progress Awards, Faculty Awards, and many best paper and honorable mention awards from conferences such as CHI and MobileHCI. Prof. Shi's recent HCI contributions includes user behavior modelling, intelligent interaction techniques for mobile phones, large displays, and VR/AR headsets. She has published more than one hundred top conference and journal papers including over 30 CHI, UIST, Ubicomp, TOCHI and IHCS papers over the most recent five years alone.

Qianying Wang, Vice President and General Manager of Technical Strategy & Innovation Platform, Lenovo, China

Qianying's position statement: We are observing more and more AI-in-the-loop applications. I would like to discuss the new mission for HCI in the Intelligence Era: from HCI to HCI (Human Centered Intelligence). Can and how HCI become adaptor or interpreter between human and AI? What are the issues brought by the low interpretability of machine learning algorithm and how to leverage existing human collaboration theories to improve human and AI relationship?

Qianying is a Lenovo Vice President, and General Manager of Technical Strategy & Innovation Platform. Stanford-educated Ph.D. in Human Computer Interaction, vice-chair of SIGCHI China Chapter. Together with her other responsibilities, she has been leading Lenovo User Experience Research and HCI team for more than 12 years and in charge of user experience of Lenovo's flagship innovative products including smart Phone, PC/Pad convergence, SmartTV, table computing, and etc. She is currently leading Lenovo Smart Education product and RD platform. In 2019, Qianying made the list of Forbes China's Top 50 "Women in Tech".

4 PANEL FORMAT AND AUDIENCE ENGAGEMENT

The panel will begin with brief (3-4 minute) position statements from each of the panelists focusing on what he/she sees as the key challenges and opportunities for Human-Centered AI research and applications. To ground the conversation, panelists will show examples of today's AI systems, and discuss design choices made with respect to how these AI systems interact with humans. There will also be a live Q&A session in which members of the audience can pose questions to the entire panel. Before, during and after the event, the audience members and panelists will be able to participate in an on-line group discussion facilitated by the moderator in Slack. We will invite the audience to use this discussion forum to inspire the community with their thoughts on the future of Human-Centered AI. We plan to report on the discussion in the panel session and in the on-line discussion to Interactions, Communications of the ACM, a special issue of a journal, or possibly a CHI 2022 submission.

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